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30 min

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Module 3 Homework

1. A drug treatment (T = treatment, C = placebo) has been randomized to 5 patients in each arm. Consider testing an increase in side effects (S = side effects, N = none). The data are

Ready to review what you've learned before taking the quiz? I'm here to help.

TP
B 3 1
N 2 4

Help me practice Let's chat

How many tables are there such that the observed counts are more extreme in favor of the alternative than the observed?

- ☐ 5
- ☐ 3
- ☒ 1
- ☐ 2
- ☐ 4

Correct
There is only one table more extreme satisfying the margins.

TP
S 4 0 4
N 1 5 5
5 5 10

Your grade
100%

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2. In a review of the author Jane Austen's work, scholars found the following relative frequencies of the words CODEan, CODEthat and "this"

an that this
0.35 0.35 0.12

An new story claimed to be Austen's was discovered with the word "an" 140 times, the word "that" 100 times and the word "this" 50 times. Are these counts consistent with Austen's traditional frequencies? How many degrees of freedom does the relevant Chi-squared test have?

- ☒ 2
- ☐ 4
- ☐ 3
- ☐ 1

Correct
This is a chi squared goodness of fit test. It requires 2 degrees of freedom.

3. In a review of the author Jane Austen's work, scholars found the following relative frequencies of the words CODEan, CODEthat and "this"

an that this
0.35 0.35 0.12

An new story claimed to be Austen's was discovered with the word "an" 140 times, the word "that" 100 times and the word "this" 50 times. Are these counts consistent with Austen's traditional frequencies? Suppose that the P-value for a chi-squared goodness of fit test turned out to be .04. At the 5% type 1 error rate what would be the conclusion?

- ☐ We conclude that the observed word distribution appears to share the same distribution as Austen's historical frequencies.
- ☐ There is insufficient evidence to make a conclusion.
- ☒ We conclude that the observed counts do not appear to share the same distribution as Austen's historical frequencies.

Correct
We reject the null hypothesis. In this case rejection implies that the word distribution does not appear to be equal to Austen's historical frequencies.

4. Two drugs, A and B, are being investigated in a randomized trial with the data are given below. Investigators would like to know if the Drug A and Drug B are equivalent in the terms of the distribution of side effects.

None
Nausea
Vomiting
Total
Drug A 80 15 5 100
Drug B 60 30 10 100

What is the chi-squared P-value?

- ☐ Around 0.05
- ☐ Around 0.10
- ☐ Closest to 0.50
- ☒ Around 0.01

Correct
1 dat <- matrix(c(80, 60, 15, 30, 5, 10), 2)
2 chisq.test(dat, correct = FALSE)

1 Pearson's Chi-squared test
2
3 data: dat
4 X-squared = 9.324, df = 2, p-value = 0.008549

5. Benford's law states that the leading digit d = 1,2,3,4,5,6,7,8,9 occurs with probability log10(d + 1) - log10(d), yielding the probabilities:

Digit 1 2 3 4 5 6 7 8 9
Probability 0.301 0.176 0.125 0.097 0.079 0.067 0.058 0.051 0.046

The following table was obtained for the leading digits from addresses randomly sampled from the phone book.

Digit 1 2 3 4 5 6 7 8 9
Count 275 183 133 111 75 66 44 46

Do the digits appear to follow Benford's law? What is the relevant P-value

- ☐ The P-value is small, we can not rule out Benford's law for this data
- ☐ The P-value is large, we can rule out Benford's law for this data
- ☒ The P-value is large, we can not rule out Benford's law for this data
- ☐ The P-value is small, we can rule out Benford's law for this data

Correct
1 dat <- c(275, 183, 133, 111, 76, 66, 66, 44, 50)
2 p <- log10((1:9) + 1) - log10(1:9)
3 chisq.test(dat, p = p, correct = FALSE)

1 Chi-squared test for given probabilities
2
3 data: dat
4 X-squared = 9.436, df = 8, p-value = 0.3069

6. Consider the table below comparing the self-reported job stress levels of three occupations. 100 people from each of three occupations were surveyed.

Job High Low
Exec 65 35
Prof 70 30
Lion Tamer 15 85

Researchers are interested in whether or not the perception of job stress differs by occupation. What can be said about the expected counts under the null hypothesis?

- ☐ The high stress expected count for the professors is larger than the high stress expected count for the lion tamers.
- ☒ All expected counts are 50
- ☐ The expected counts can't be obtained for the problem as stated.
- ☐ The margins of the expected counts do not agree with the margins of the observed counts

Correct
There were 50% high and low stress disregarding profession

1 dat <- matrix(c(65, 70, 15, 35, 30, 85), 3)
2 colSums(dat)/sum(dat)

1 [1] 0.5 0.5

So, all expected counts are 50 since all of the row totals are 100